林亦波

长聘副教授 ◇ 设计自动化与计算系统系 ◇ 集成电路学院 ◇ 北京大学

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研究方向

面向超大规模集成电路设计自动化的建模和优化、深度学习及其应用、异构计算

教育经历

德克萨斯大学奥斯汀分校 2013 年 9 月 – 2018 年 5 月
博士学位, 电子与计算机工程系

上海交通大学 2009 年 9 月 – 2013 年 7 月
学士学位, 微电子学院

工作经历

北京大学 (Peking University) 2025 年 8 月至今
长聘副教授

北京大学 (Peking University) 2019 年 6 月 – 2025 年 7 月
助理教授
集成电路学院设计自动化与计算系统系 (自 2021 年 11 月)
信息科学技术学院高能效计算与应用中心

德克萨斯大学奥斯汀分校 (University of Texas at Austin) 2018 年 7 月 – 2019 年 6 月
博士后

授课经历

主讲	AI 赋能集成电路设计实践	研究生课程, 2026 年
主讲	集成电路工程算法	研究生课程, 2023-2026 年
主讲	设计自动化与计算系统	研究生课程, 2022-2024 年
主讲	芯片设计自动化与智能优化	本科生课程, 2021-2025 年
主讲	计算概论 B	本科生课程, 2020-2022 年

奖项及荣誉

DAC“40 岁以下优秀创新者奖”	DAC	2026
最佳论文提名 (13/544)	DAC	2026
最佳论文提名 (×3) (12/145)	ISEDA	2026
最佳论文提名 (19/266)	ICCAD	2025 年
最佳论文提名 (7/420)	DAC	2025 年
荣誉提名论文 (×2)	ISEDA	2025 年
最佳论文提名 (13/168)	ASPDAC	2025 年
最佳论文提名 (10/195)	ICCAD	2024 年
最佳论文 (6/151) & 荣誉提名论文	ISEDA	2024 年

集成电路青年科技奖 (每年仅 1 位)	中国计算机学会容错计算专委	2023 年
首届最佳审稿人奖	ICCAD	2023 年
最佳论文 (4/205)	DATE	2023 年
最佳论文 (4/249)	DATE	2022 年
最佳论文提名 (6/132)	ICCAD	2022 年
最佳论文 (2/3495, 4 年)	TCAD	2021 年
最佳论文	ISPD	2020 年
最佳论文提名 (12/86)	ASPDAC	2020 年
最佳论文 (1/201) & 提名 (5/201)	DAC	2019 年
最佳论文提名	ISPD	2019 年
首届最佳论文	Integration, the VLSI Journal	2018 年
Franco Cerrina Memorial 最佳学生论文	SPIE	2016 年
A. Richard Newton Young Student Fellow	DAC	2014 年

学术服务

执行委员会成员

- 大会共同主席, ACM/IEEE International Symposium on Machine Learning for CAD (MLCAD), 2025 年–2026 年
- 程序共同主席, ACM/IEEE International Symposium on Machine Learning for CAD (MLCAD), 2024 年
- 圆桌论坛主席, IEEE International Symposium of Electronics Design Automation (ISEDA), 2023 年–2024 年
- 财务主席, ACM/IEEE Workshop on Machine Learning for CAD (MLCAD), 2021 年–2023 年

期刊编委

- 副编辑,IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2024 年至今
- 副编辑，ACM Transaction on Design Automation of Electronic Systems (TODAES), 2024 年至今
- 副编辑，Elsevier Integration, the VLSI Journal (Integration), 2024 年至今
- 客座编辑, ACM Transaction on Design Automation of Electronic Systems (TODAES) Special Issue on MLCAD, 2024 年
- 客座编辑, ACM Transaction on Design Automation of Electronic Systems (TODAES) Special Issue on MLCAD, 2022 年

赛题主席

- 赛题主席，集成电路 EDA 设计精英挑战赛，2021 年–2023 年

技术程序委员会成员

- ACM/IEEE Design Automation Conference (DAC): 2020
- IEEE/ACM International Conference on Computer-Aided Design (ICCAD): 2018 – 2021, 2023
- Design, Automation and Test in Europe Conference (DATE): 2025
- IEEE International Conference on Computer Design (ICCD): 2019

- IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC): 2021 – 2022
- ACM International Symposium on Physical Design (ISPD): 2020, 2025
- ACM/IEEE Workshop on Machine Learning for CAD (MLCAD): 2021
- ACM Great Lakes Symposium on VLSI (GLVLSI): 2024
- Workshop on Synthesis And System Integration of Mixed Information technologies (SASIMI): 2021
- IEEE Electron Devices Technology and Manufacturing Conference (EDTM): 2021
- IEEE International Conference on Artificial Intelligence Circuits and Systems (AICAS): 2022.

期刊审稿人

- IEEE Transaction on Computer-Aided Design of Integrated Circuits and Systems (TCAD)
- IEEE Transactions on Computers (TC)
- ACM Transaction on Design Automation of Electronic Systems (TODAES)
- SPIE Journal of Micro/Nanolithography, MEMS, and MOEMS (JM3)
- Elsevier, Integration, the VLSI Journal (Integration)

其他志愿服务

- ACM SIGDA 网站管理负责人，2021 年至今

10 篇代表性论文

按时间倒序排列。标 * 表示通讯作者。

1. Yifan Chen, Zaiwen Wen, Yun Liang and **Yibo Lin***, "Stronger Mixed-Size Placement Backbone Considering Second-Order Information", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), San Francisco, CA, Oct 29-31, 2023.
2. Haoyi Zhang, Xiaohan Gao, Haoyang Luo, Jiahao Song, Xiyuan Tang, Junhua Liu, **Yibo Lin***, Runsheng Wang and Ru Huang, "SAGERoute: Synergistic Analog Routing Considering Geometric and Electrical Constraints with Manual Design Compatibility", IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Antwerp, Belgium, Apr 17-19, 2023. (**Best Paper Award**)
3. Jing Mai, Jiarui Wang, Zhixiong Di and **Yibo Lin***, "Multi-Electrostatic FPGA Placement Considering SLICEL-SLICEM Heterogeneity, Clock Feasibility, and Timing Optimization", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Sep, 2023.
4. Zizheng Guo, Tsung-Wei Huang and **Yibo Lin***, "Accelerating Static Timing Analysis using CPU-GPU Heterogeneous Parallelism", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Jun, 2023.
5. Qipan Wang, Xiaohan Gao, **Yibo Lin***, Runsheng Wang and Ru Huang, "DeePEB: A Neural Partial Differential Equation Solver for Post Exposure Baking Simulation in Lithography", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), San Diego, CA, Nov 01-03, 2022. (**Best Paper Nomination**)
6. Zizheng Guo and **Yibo Lin***, "Differentiable-Timing-Driven Global Placement", ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jul 10-14, 2022.

7. Siting Liu, Peiyu Liao, Zhitang Chen, Wenlong Lv, **Yibo Lin*** and Bei Yu*, "FastGR: Global Routing on CPU-GPU with Heterogeneous Task Graph Scheduler", IEEE/ACM Proceedings Design, Automation and Test in Europe (DATE), Antwerp, Belgium, Mar 14-23, 2022. (**Best Paper Award**)
8. Zizheng Guo, Jing Mai and **Yibo Lin***, "Ultrafast CPU/GPU Kernels for Density Accumulation in Placement", ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Dec 05-09, 2021.
9. Zizheng Guo, Tsung-Wei Huang and **Yibo Lin***, "A Provably Good and Practically Efficient Algorithm for Common Path Pessimism Removal in Large Designs", ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Dec 05-09, 2021.
10. **Yibo Lin***, Zixuan Jiang, Jiaqi Gu, Wuxi Li, Shounak Dhar, Haoxing Ren, Brucek Khailany and David Z. Pan, "DREAMPlace: Deep Learning Toolkit-Enabled GPU Acceleration for Modern VLSI Placement", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Apr, 2021. (**Best Paper Award**)

发明专利列表

[P33] **林亦波**, 江循, 吴恒, 王润声, "基于双面晶体管的拥塞线长同时感知的标准单元选型方法", 中国专利, CN202511719692X, Feb 17, 2026. (submitted)

[P32] **林亦波**, 刘海川, 郭资政, 张作栋, 王兆卿, "集成电路增量时序分析方法、装置和计算机设备", 中国专利, CN2025114363723, Jan 30, 2026. (submitted)

[P31] **林亦波**, 江循, 吴恒, 王润声, "一种基于双面晶体管的信号传输单元插入方法", 中国专利, CN2025114574160, Jan 13, 2026. (submitted)

[P30] 余备, 蒲渊, 纪宇豪, 张作栋, 郭资政, **林亦波**, 王兆卿, "缓冲器插入方法和装置", 中国专利, CN2025113264735, Dec 26, 2025. (submitted)

[P29] **林亦波**, 杜宇凡, 郭资政, 张作栋, 王兆卿, "一种集成电路数字后端物理优化方法、设备、介质及产品", 中国专利, CN2025116499840, Dec 12, 2025. (submitted)

[P28] **林亦波**, 麦景, 赵春源, 张作栋, 王兆卿, 王润声, "集成电路合法化方法和装置", 中国专利, CN2025111769595, Dec 02, 2025. (submitted)

[P27] **林亦波**, 郭铠荣, "一种双面三维堆叠晶体管标准单元版图生成方法", 中国专利, CN2025111087257, Nov 21, 2025. (submitted)

[P26] **林亦波**, 喻文健, 张昊懿, 黄杰辰, "一种实时考虑模拟集成电路版图寄生的布线方法", 中国专利, CN2025111878075, Nov 18, 2025. (submitted)

[P25] **林亦波**, 赵春源, 王嘉睿, 江循, 楼锦程, 张作栋, 王兆卿, 王润声, "布线轨道分配方法、装置、计算机设备和可读存储介质", 中国专利, CN202510862644X, Nov 18, 2025. (submitted)

[P24] **林亦波**, 郭铠荣, "一种面向多行标准单元的层次化晶体管布局方法", 中国专利, CN2025106118325, Aug 22, 2025.

[P23] **林亦波**, 杜宇凡, 郭资政, 张作栋, 王润声, "芯片元件位置和尺寸调整方法", 中国专利, CN2025102641639, Jun 20, 2025. (submitted)

[P22] **林亦波**, 郭资政, 张作栋, 王润声, 黄如, "集成电路延迟确定方法、装置、设备、介质和产品", 中国专利, CN202411597634X, Feb 18, 2025. (submitted)

[P21] 余备, 蒲渊, **林亦波**, "一种基于时序优化的 Macro 布局增量优化方法", 中国专利, CN2024109403891, Nov 29, 2024. (submitted)

[P20] **林亦波**, 赵春源, 张作栋, "一种 3-bend 模式布线方法", 中国专利, CN2024108984500, Nov 29, 2024. (submitted)

[P19] **林亦波**, 王启盼, 王润声, 黄如, "一种集成芯片稳态热仿真方法", 中国专利, CN2024108520949, Oct 25, 2024. (submitted)

[P18] 吴恒, 卢浩然, 江循, 孙嘉诚, 王润声, **林亦波**, 黄如, "半导体结构的制备方法、半导体结构、器件及设备", 中国专利, CN2024101785421, May 24, 2024.

[P17] **林亦波**, 郭资政, 张作栋, 江循, 王润声, 黄如, "电路门级逻辑仿真方法、装置、计算机设备和存储介质", 中国专利, CN2023112735027, Jan 09, 2024.

[P16] 余备, 廖培宇, **林亦波**, "一种基于 Moreau Envelope 近似线长模型的大规模解析布局方法", 中国专利, CN2023111300780, Nov 28, 2023. (submitted)

[P15] 王润声, 张作栋, **林亦波**, 黄如, "一种基于解析模型的晶体管老化应力计算方法", 中国专利, CN202310843291X, Oct 13, 2023.

[P14] **林亦波**, 高笑涵, 张昊懿, 王润声, 黄如, "一种多驱动能力的集成电路标准单元版图迁移的方法", 中国专利, CN2023101249631, Mar 28, 2023.

[P13] **林亦波**, 王启盼, 王润声, 黄如, "一种集成电路微带线传输线自动化分析设计方法", 中国专利, CN2023100496576, Mar 17, 2023.

[P12] **林亦波**, 郭资政, 谷丰, "一种 GPU 加速构建最小直角斯坦纳树的芯片布线方法", 中国专利, CN2022112858018, Jan 03, 2023.

[P11] **林亦波**, 高笑涵, 张昊懿, 王润声, 黄如, "可处理电学和几何约束的模拟电路布线自动化方法及系统", 中国专利, CN2022114229951, Dec 20, 2022.

[P10] 郭资政, **林亦波**, 黄琮蔚, "一种集成电路静态时序分析中的路径分析方法", 中国专利, CN2021103772507, Oct 18, 2022.

[P9] **林亦波**, 郭资政, "一种可微分时序驱动的芯片布局优化方法", 中国专利, CN2022107930171, Aug 05, 2022.

[P8] **林亦波**, 张昊懿, 高笑涵, 王润声, 黄如, "一种用于模拟电路版图布线的交互式编辑方法及工具", 中国专利, CN2022100363194, May 17, 2022.

[P7] **林亦波**, 麦景, "基于多电场模型的时钟驱动 FPGA 芯片全局布局方法", 中国专利, CN2022102058942, Apr 12, 2022.

[P6] **林亦波**, 张作栋, 郭资政, 王润声, 黄如, "一种老化及涨落感知的动态时序分析方法", 中国专利, CN2021115414669, Apr 05, 2022.

[P5] **林亦波**, 张作栋, 郭资政, 王润声, 黄如, "一种基于事件传播的动态时序分析方法", 中国专利, CN2021109930951, Dec 24, 2021.

[P4] **林亦波**, 郭资政, 黄琮蔚, "一种 GPU 加速计算的集成电路无悲观路径分析方法", 中国专利, CN2021110703249, Dec 24, 2021.

[P3] 麦景, 郭资政, **林亦波**, "一种集成电路设计中器件密度分布的计算方法", 中国专利, CN2021105506486, Aug 27, 2021.

[P2] 高笑涵, **林亦波**, 刘鸣杰, 潘志刚, "一种交互式模拟电路版图编辑方法及系统", 中国专利, CN2021101747163, Jun 18, 2021.

[P1] 郭资政, 黄琮蔚, **林亦波**, "一种 GPU 加速计算的集成电路静态时序分析方法", 中国专利, CN2020111436325, Jan 22, 2021.

著作及论文列表

书籍章节

[B3] **Yibo Lin**, Zizheng Guo and Jing Mai, "Deep Learning Framework for Placement", Machine Learning Applications in Electronic Design Automation, Springer, 2023, edited by Haoxing Ren and Jiang Hu. (**Invited Book Chapter**)

[B2] Haoyu Yang, **Yibo Lin** and Bei Yu, "Machine Learning for Mask Synthesis and Verification", Machine Learning Applications in Electronic Design Automation, Springer, 2023, edited by Haoxing Ren and Jiang Hu. (**Invited Book Chapter**)

会议及期刊论文 (标 * 表示通讯作者)

论文成果包括: DAC (37 篇), ICCAD (32 篇), IEEE TCAD (35 篇), DATE (21 篇), ...

[J252] Zhenkun Lin, **Yibo Lin**, Genggeng Liu* and Gang Du, "URoute: Universal Routability Prediction", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2026. (accepted)

[J251] Xu He*, Yushan Wang, Chenjing Yang, Renjun Zhao, **Yibo Lin**, Peiyu Liao, Bei Yu, Yao Wang, Chang Liu and Yang Guo, "Efficient Timing Prediction and Optimization Using Derivable Gradient Boosting Machine Model at Placement Stage", ACM Transactions on Design Automation of Electronic Systems (TODAES), 2026. (accepted)

[J250] Chunyuan Zhao, Zizheng Guo, Zuodong Zhang and **Yibo Lin***, "GAP-LA: GPU-Accelerated Performance-Driven Layer Assignment", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2026. (accepted)

[J249] Xufeng Yao, Jiaxi Jiang, Yuxuan Zhao, Peiyu Liao, **Yibo Lin** and Bei Yu*, "EvoPlace: Evolution of Optimization Algorithms for Global Placement via Large Language Models", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2026. (accepted)

[J248] Tianxiang Zhu, Qipan Wang, **Yibo Lin*** and Runsheng Wang*, "M2MOR: Multi-scale Thermo-mechanical Coupling Simulation of Chiplet Heterogeneous Integration Systems Based on Model Order Reduction", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2026. (accepted)

[J247] Haoyi Zhang, Jiechen Huang, Bingyang Liu, Wenjian Yu*, **Yibo Lin*** and Runsheng Wang, "APEC-Route: Guiding Analog Routing with On-the-Fly Parasitic Extraction Based on a Neural Capacitance Model", ACM Transactions on Design Automation of Electronic Systems (TODAES), 2026. (accepted)

[J246] Yifan Chen, Zixiao Wang, Zhe Liu, Ke Xu, Meng Li, Bei Yu, Runsheng Wang and **Yibo Lin***, "SoftClay3D: Exploring Intra-Module Partitioning with True-3D Floorplan and Placement Engines", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2026. (accepted)

[C245] Mengjia Dai, Hongduo Liu, Yuhao Ji, Xiaojing Su, **Yibo Lin** and Bei Yu*, "Soft-Constrained Triple Patterning Layout Decomposition", IEEE International Conference on Computer Design (ICCD), Hong Kong, China, Nov 16-18, 2026. (accepted)

[C244] Xiang Li, Zuodong Zhang and **Yibo Lin***, "TAT: Timing-Aware Steiner Tree Generation for Buffer Insertion", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), San Jose, CA, Nov 8-12, 2026. (accepted)

[C243] Yuxi Lin, **Yibo Lin**, Zepeng Li, Weijie Fang, Guangyong Chen and Genggeng Liu*, "Clock Tree Synthesis with Cross-Tier Skew and Wirelength Co-Optimization for Face-to-Face Hybrid Bonding 3D ICs", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), San Jose, CA, Nov 8-12, 2026. (accepted)

[C242] Zhenkun Lin, Taoyu Zhou, Qipan Wang, Tianxiang Zhu, **Yibo Lin**, Meng Li, Runsheng Wang, Yandong He and Gang Du*, "VULCAN: Unified Multi-Scale Chip Thermal Simulation System for Design-Technology Co-Optimization", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), San Jose, CA, Nov 8-12, 2026. (accepted)

[C241] Jingxin Wang, Sinian Zhang, Yun Liang, **Yibo Lin**, Pengpeng Ren, Runsheng Wang and Weikang Qian*, "GUALO: A Generalizable Uncertainty-Aware AI Agent for Logic Optimization", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), San Jose, CA, Nov 8-12, 2026. (accepted)

[C240] Jincheng Lou, Ruohan Xu, Jiecheng Ma, Boyang Wu, Yuhao Ji, Chunyuan Zhao, Xinyu Qu, Yuxiang Zhao, Runzhe Tao, Bei Yu, Zhixiong Di and **Yibo Lin***, "LEGO 2.0: A Unified Agentic Platform for Digital Chip Design and Optimization", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), San Jose, CA, Nov 8-12, 2026. (accepted)

[C239] Weijian Fan, Zhiyuan Zhao, Jinpeng Wang, Haoyi Zhang, Jincheng Lou, Runsheng Wang, Xiyuan Tang and **Yibo Lin***, "VinePlace: Compact Analog and Mixed-Signal Placement via Core-Centered Vine Representation", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), San Jose, CA, Nov 8-12, 2026. (accepted)

[C238] Yuhao Ji, Siying Yu, Yunqi Shi, Fangzhou Liu, Yuxuan Zhao, Jiaxi Jiang, Zizheng Guo, Chao Qian, **Yibo Lin** and Bei Yu*, "HeteroOpt3D: Timing Optimization for Heterogeneous Hybrid-bonding-based Face-to-Face 3D ICs", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), San Jose, CA, Nov 8-12, 2026. (accepted)

[C237] Kairong Guo, Haoyi Zhang, Haichuan Liu, Chunyuan Zhao and **Yibo Lin***, "ScoutCell: Sensitivity-Guided Coupling Optimization for Timing-Driven Standard Cell Synthesis", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), San Jose, CA, Nov 8-12, 2026. (accepted)

[C236] Zixuan Pan, Zizheng Guo, Yufan Du, Runsheng Wang and **Yibo Lin***, "Disentangled Differentiable Timing-Power Co-Optimization with Quad-Gradient Gate sizing", ACM/IEEE Design Automation Conference (DAC), Long Beach, CA, Jul 26-29, 2026. (**Best Paper Nomination**)

[C235] Yi Ren, Yukun Wang, Xiang Meng, Guoyao Cheng, Baokang Peng, Lining Zhang, **Yibo Lin**, Runsheng Wang and Guangyu Sun*, "CellE: Automated Standard Cell Library Extension via Equality Saturation", ACM/IEEE Design Automation Conference (DAC), Long Beach, CA, Jul 26-29, 2026.

[C234] Jing Mai, Yi-Chen Lu, **Yibo Lin** and Haoxing Ren, "LATTE: Legality-Assured Differentiable Timing-Driven Detailed Placement", ACM/IEEE Design Automation Conference (DAC), Long Beach, CA, Jul 26-29, 2026.

[C233] Wenbo An, Kairong Guo, Haoyi Zhang and **Yibo Lin***, "EPiCell: Electro-Physical Co-Modeling for Standard Cell PPA Prediction", ACM/IEEE Design Automation Conference (DAC), Long Beach, CA, Jul 26-29, 2026.

[C232] Yuhao Ji, Yunqi Shi, Tianshu Hou, Yuxuan Zhao, Chunyuan Zhao, Peiyu Liao, Zizheng Guo, Chao Qian, **Yibo Lin** and Bei Yu*, "Integrated Timing-driven Placement for Hybrid-Bonding-based Face-to-Face 3D ICs", ACM/IEEE Design Automation Conference (DAC), Long Beach, CA, Jul 26-29, 2026.

[C231] Kexing Zhou, Youwei Zhuo, **Yibo Lin**, Weikang Qian, Pengpeng Ren and Yun Liang*, "Activity-Aware Partitioning for Effective Multi-Threaded Event-Driven RTL Simulation", ACM/IEEE Design Automation Conference (DAC), Long Beach, CA, Jul 26-29, 2026.

[C230] Jiechen Huang, Hector R. Rodriguez, Dingcheng Yang, Zuochang Ye, **Yibo Lin** and Wenjian Yu*, "AttentionCap: Transformer Based Capacitance Matrix Learning Toward Full-Chip Extraction", ACM/IEEE Design Automation Conference (DAC), Long Beach, CA, Jul 26-29, 2026.

[C229] Haoyi Zhang, Weijian Fan, Xiaohan Gao, Bingyang Liu, Runsheng Wang and **Yibo Lin***, "PANDA: An LLM-Enhanced Performance-Driven Analog Design Framework Bridging Design Intent and Layout Generation", ACM/IEEE Design Automation Conference (DAC), Long Beach, CA, Jul 26-29, 2026. (**Invited Paper**)

[C228] Haoyi Zhang, Kairong Guo, Bojie Zhang, **Yibo Lin*** and Runsheng Wang, "FusionCell: Cross-Attentive Fusion of Layout Geometry and Netlist Topology for Standard-Cell Performance Prediction", International Conference on Machine Learning (ICML), Seoul, Korea, Jul 6-11, 2026.

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========== 以下为博士及博士后期间发表内容 ==========

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会议及期刊论文 (标 * 表示通讯作者)

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[J44] **Yibo Lin**, Meng Li, Yuki Watanabe, Taiki Kimura, Tetsuaki Matsunawa, Shigeki Nojima and David Z. Pan*, "Data Efficient Lithography Modeling with Transfer Learning and Active Data Selection", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Oct, 2019.

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[C41] Wei Ye, Mohamed Baker Alawieh, **Yibo Lin** and David Z. Pan*, "LithoGAN: End-to-End Lithography Modeling with Generative Adversarial Networks", ACM/IEEE Design Automation Conference (DAC), Las Vegas, NV, Jun 2-6, 2019. (**Best Paper Nomination**)

[C40] Biying Xu, **Yibo Lin**, Xiyuan Tang, Shaolan Li, Linxiao Shen, Nan Sun and David Z. Pan*, "WellGAN: Generative-Adversarial-Network-Guided Well Generation for Analog/Mixed-Signal Circuit Layout", ACM/IEEE Design Automation Conference (DAC), Las Vegas, NV, Jun 2-6, 2019.

[C39] Mohamed Baker Alawieh, **Yibo Lin**, Zaiwei Zhang, Meng Li, Qixing Huang and David Z. Pan*, "GAN-SRAF: Sub-Resolution Assist Feature Generation Using Conditional Generative Adversarial Networks", ACM/IEEE Design Automation Conference (DAC), Las Vegas, NV, Jun 2-6, 2019.

[C38] **Yibo Lin**, Zhao Song and Lin F. Yang*, "Towards a Theoretical Understanding of Hashing-Based Neural Nets", International Conference on Artificial Intelligence and Statistics (AISTATS), Okinawa, Japan, Apr 16-18, 2019.

[C37] Biying Xu, Shaolan Li, Chak-Wa Pui, Derong Liu, Linxiao Shen, **Yibo Lin**, Nan Sun and David Z. Pan*, "Device Layer-Aware Analytical Placement for Analog Circuits", ACM International Symposium on Physical Design (ISPD), San Francisco, CA, Apr 14-17, 2019. (**Best Paper Nomination**)

[J36] Ying Chen, **Yibo Lin**, Tianyang Gai, Yajuan Su*, Yayi Wei* and David Z. Pan, "Semi-Supervised Hotspot Detection with Self-Paced Multi-Task Learning", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Apr, 2019.

[C35] Wei Ye, Mohamed Baker Alawieh, Meng Li, **Yibo Lin** and David Z. Pan*, "Litho-GPA: Gaussian Process Assurance for Lithography Hotspot Detection", IEEE/ACM Proceedings

Design, Automation and Test in Europe (DATE), Florence, Italy, Mar 25-29, 2019.

[J34] Jing Chen, **Yibo Lin**, Yufeng Guo, Maolin Zhang, Mohamed Baker Alawieh and David Z. Pan*, "Lithography Hotspot Detection Using a Double Inception Module Architecture", Journal of Micro/Nanolithography, MEMS, and MOEMS (JM3), Mar, 2019.

[C33] Ying Chen, **Yibo Lin**, Tianyang Gai, Yajuan Su*, Yayi Wei* and David Z. Pan, "Semi-Supervised Hotspot Detection with Self-Paced Multi-Task Learning", IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Tokyo, Japan, Jan 21-24, 2019.

[C32] Wei Ye, Mohamed Baker Alawieh, **Yibo Lin** and David Z. Pan*, "Tackling Signal Electromigration with Learning-Based Detection and Multistage Mitigation", IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Tokyo, Japan, Jan 21-24, 2019.

[C31] Wei Ye, **Yibo Lin**, Meng Li, Qiang Liu and David Z. Pan*, "LithoROC: Lithography Hotspot Detection with Explicit ROC Optimization", IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Tokyo, Japan, Jan 21-24, 2019. (**Invited Paper**)

[C30] **Yibo Lin**, Mohamed Baker Alawieh, Wei Ye and David Z. Pan*, "Machine Learning for Yield Learning and Optimization", IEEE International Test Conference (ITC), Phoenix, Arizona, Oct, 2018. (**Invited Paper**)

[J29] **Yibo Lin**, Bei Yu, Meng Li and David Z. Pan*, "Layout Synthesis for Topological Quantum Circuits with 1D and 2D Architectures", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Aug, 2018.

[J28] Meng Li, Bei Yu, **Yibo Lin**, Xiaoqing Xu, Wuxi Li and David Z Pan, "A practical split manufacturing framework for trojan prevention via simultaneous wire lifting and cell insertion", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Jul, 2018.

[C27] Jiong Zhang, **Yibo Lin**, Zhao Song and Inderjit S Dhillon*, "Learning Long Term Dependencies via Fourier Recurrent Units", International Conference on Machine Learning (ICML), Stockholm, Sweden, Jun 10-15, 2018.

[J26] Xiaoqing Xu, **Yibo Lin**, Meng Li, Tetsuaki Matsunawa, Shigeki Nojima, Chikaaki Kodama, Toshiya Kotani and David Z. Pan*, "Subresolution Assist Feature Generation With Supervised Data Learning", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Jun, 2018.

[J25] **Yibo Lin**, Bei Yu, Xiaoqing Xu, Jhih-Rong Gao, Natarajan Viswanathan, Wen-Hao Liu, Zhuo Li, Charles J Alpert and David Z. Pan*, "MrDP: Multiple-row detailed placement of heterogeneous-sized cells for advanced nodes", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Jun, 2018.

[J24] Wuxi Li, **Yibo Lin**, Meng Li, Shounak Dhar and David Z. Pan*, "UTPlaceF 2.0: A High-Performance Clock-Aware FPGA Placement Engine", ACM Transactions on Design Automation of Electronic Systems (TODAES), Jun, 2018.

[C23] **Yibo Lin**, Yuki Watanabe, Taiki Kimura, Tetsuaki Matsunawa, Shigeki Nojima, Meng Li and David Z. Pan*, "Data Efficient Lithography Modeling with Residual Neural Networks and Transfer Learning", ACM International Symposium on Physical Design (ISPD), Monterey, CA, Mar 25-28, 2018.

[C22] Meng Li, Bei Yu, **Yibo Lin**, Xiaoqing Xu, Wuxi Li and David Z. Pan*, "A Practical Split Manufacturing Framework for Trojan Prevention via Simultaneous Wire Lifting and Cell Insertion", IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Jeju, Korea, Jan 22-25, 2018.

[C21] Che-Lun Hsu, Shaofeng Guo, **Yibo Lin**, Xiaoqing Xu, Meng Li, Runsheng Wang, Ru Huang and David Z Pan, "Layout-dependent aging mitigation for critical path timing", IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Jeju, Korea, Jan 22-25, 2018.

[C20] **Yibo Lin**, Peter Debacker, Darko Trivkovic, Ryoung-han Kim, Praveen Raghavan and David Z. Pan*, "Patterning Aware Design Optimization of Selective Etching in N5 and Beyond", IEEE International Conference on Computer Design (ICCD), Boston, MA, Nov 5-8, 2017.

[C19] **Yibo Lin**, Xiaoqing Xu, Jiaojiao Ou and David Z Pan, "Machine learning for mask/wafer hotspot detection and mask synthesis", Photomask Technology, Oct 16, 2017. (**Invited Paper**)

[J18] **Yibo Lin**, Bei Yu and David Z. Pan*, "High performance dummy fill insertion with coupling and uniformity constraints", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Sep, 2017.

[C17] Wei Ye, **Yibo Lin**, Xiaoqing Xu, Wuxi Li, Yiwei Fu, Yongsheng Sun, Canhui Zhan and David Z. Pan*, "Placement Mitigation Techniques for Power Grid Electromigration", IEEE International Symposium on Low Power Electronics and Design (ISLPED), Taipei, Jul 24-26, 2017.

[J16] **Yibo Lin**, Bei Yu, Biying Xu and David Z. Pan*, "Triple patterning aware detailed placement toward zero cross-row middle-of-line conflict", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Jul, 2017.

[J15] Xiaoqing Xu, **Yibo Lin**, Meng Li, Jiaojiao Ou, Brian Cline and David Z. Pan*, "Redundant local-Loop insertion for unidirectional routing", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), Jul, 2017.

[C14] Xiaoqing Xu, **Yibo Lin**, Vinicius Livramento and David Z. Pan*, "Concurrent Pin Access Optimization for Unidirectional Routing", ACM/IEEE Design Automation Conference (DAC), Austin, TX, Jun 18-22, 2017.

[J13] **Yibo Lin**, Bei Yu, Yi Zou, Zhuo Li, Charles J Alpert and David Z. Pan*, "Stitch aware detailed placement for multiple e-beam lithography", Integration, the VLSI Journal, Jun, 2017. (**Best Paper Award**)

[J12] **Yibo Lin**, Xiaoqing Xu, Bei Yu, Ross Baldick and David Z. Pan*, "Triple/quadruple patterning layout decomposition via linear programming and iterative rounding", Journal of Micro/Nanolithography, MEMS, and MOEMS (JM3), Jun, 2017.

[C11] Jiaojiao Ou, Bei Yu, Xiaoqing Xu, Joydeep Mitra, **Yibo Lin** and David Z. Pan*, "DSAR: DSA aware routing with simultaneous DSA guiding pattern and double patterning assignment", ACM International Symposium on Physical Design (ISPD), Portland, OR, Mar 19-22, 2017.

[C10] **Yibo Lin**, Bei Yu, Xiaoqing Xu, Jhih-Rong Gao, Natarajan Viswanathan, Wen-Hao Liu, Zhuo Li, Charles J Alpert and David Z. Pan*, "MrDP: Multiple-row detailed placement

of heterogeneous-sized cells for advanced nodes", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Austin, TX, Nov 7-10, 2016.

[C9] Yudong Tao, Changhao Yan*, **Yibo Lin**, Sheng-Guo Wang, David Z. Pan and Xuan Zeng, "A novel unified dummy fill insertion framework with SQP-based optimization method", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Austin, TX, Nov 7-10, 2016.

[C8] **Yibo Lin**, Bei Yu and David Z. Pan*, "Detailed placement in advanced technology nodes: a survey", IEEE International Conference on Solid-State and Integrated Circuit Technology (ICSICT), Hangzhou, China, Oct 25-28, 2016. (**Invited Paper**)

[J7] Bei Yu, Xiaoqing Xu, Subhendu Roy, **Yibo Lin**, Jiaojiao Ou and David Z. Pan*, "Design for manufacturability and reliability in extreme-scaling VLSI", Science China Information Sciences, May, 2016. (**Invited Paper**)

[C6] **Yibo Lin**, Xiaoqing Xu, Bei Yu, Ross Baldick and David Z. Pan*, "Triple/quadruple patterning layout decomposition via novel linear programming and iterative rounding", Proceedings of SPIE, San Jose, CA, Feb 21-25, 2016. (**Best Student Paper Award**)

[C5] **Yibo Lin**, Bei Yu, Yi Zou, Zhuo Li, Charles J Alpert and David Z. Pan*, "Stitch aware detailed placement for multiple e-beam lithography", IEEE/ACM Asia and South Pacific Design Automation Conference (ASPDAC), Macau, China, Jan 25-28, 2016.

[C4] **Yibo Lin**, Bei Yu, Biying Xu and David Z. Pan*, "Triple patterning aware detailed placement toward zero cross-row middle-of-line conflict", IEEE/ACM International Conference on Computer-Aided Design (ICCAD), Austin, TX, Nov 2-6, 2015.

[C3] **Yibo Lin**, Bei Yu and David Z. Pan*, "High performance dummy fill insertion with coupling and uniformity constraints", ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jun 7-11, 2015.

[C2] David Z. Pan*, Lars Liebmann, Bei Yu, Xiaoqing Xu and **Yibo Lin**, "Pushing multiple patterning in sub-10nm: are we ready?", ACM/IEEE Design Automation Conference (DAC), San Francisco, CA, Jun 7-11, 2015. (**Invited Paper**)

[J1] Bei Yu, Xiaoqing Xu, Jhih-Rong Gao, **Yibo Lin**, Zhuo Li, Charles Alpert and David Z. Pan*, "Methodology for standard cell compliance and detailed placement for triple patterning lithography", IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD), May, 2015.